

SIH26143

Attribution Engine, Novelty & System Architecture

Evidence-Based Vessel Attribution • Novelty Positioning • System Interfaces

Purpose

A focused research brief covering the attribution/intelligence layer of SIH26143 — the module that takes candidate vessel evidence (from AIS/backtracking) and the spill mask/geometry (from satellite detection) and turns them into a ranked, explainable list of probable source vessels. Written to move from decision to a working scoring module within roughly one day, consistent with what the detection and AIS/GIS modules are building in parallel.

Decision Recommendation

Item	Decision
Primary attribution method	Weighted multi-evidence scoring (rule-based, explainable)
Stretch method	Bayesian/likelihood refinement of the same evidence, if time allows
Core evidence signals	Distance, time consistency, heading, speed, vessel type, track continuity, environmental consistency
Primary novelty claim	Low-cost, India-focused, explainable multi-evidence attribution with drift-aware backtracking
System role	Consumes Martin's spill output + Jitaan's AIS/backtracking output; produces ranked vessel list for Rudra's dashboard
Main failure mode	Overclaiming certainty — must present scores as probabilistic, not legal proof

Research status: built on top of the team's already-verified satellite (Martin) and environmental/prior-art (Rudra) research. Claims about existing systems (CleanSeaNet, Cerulean) and papers are drawn from sources already checked by the team; this document adds the attribution-model and architecture layer on top.

Executive Summary

SIH26143 needs more than detection plus AIS lookup — it needs a defensible answer to *"why this vessel and not that one."* The satellite module (Martin) will hand off a spill polygon, centroid, timestamp and confidence. The AIS/GIS module (Jitaan) will hand off a shortlist of candidate vessels with spatial, temporal and motion evidence, plus a backtracked source corridor. My module sits between those outputs and the dashboard (Rudra): it aggregates the evidence per candidate vessel into a single explainable confidence score and ranks them.

The two things that matter most for a hackathon jury are (1) a scoring approach we can actually finish and explain on stage, and (2) a novelty claim that survives push-back. Both are addressed below: a weighted scoring model is recommended over Bayesian/graph approaches for implementation reasons, and the novelty claim is built around explainable multi-evidence fusion plus drift-aware backtracking, not "first to combine satellite and AIS" (existing systems already do that).

1. Pipeline Ownership

The end-to-end system, and where my module sits in it:

Pipeline stage	Owner
Satellite imagery	Martin
Oil-spill detection → spill mask	Martin
Spill polygon + centroid + timestamp + confidence	Martin
Wind/current context → backtracked source corridor	Jitaan
AIS trajectories → candidate vessel shortlist + raw evidence	Jitaan
Evidence aggregation → per-vessel confidence score	Om (this module)
Ranked, explainable vessel attribution output	Om (this module)
Map + sidebar + candidate ranking display	Rudra

I need to understand the rest of the pipeline well enough to design clean interfaces around it, but the detection method, AIS parsing and backtracking math are not being re-researched here — see Martin's and Jitaan's briefs for that depth.

2. Evidence Variables for Attribution

For each vessel candidate, the following signals increase or decrease the likelihood that it is the spill source. This list is the direct input to the scoring model in Section 4.

Evidence signal	Why it moves the score
Distance to inferred origin	Closer to the backtracked source region → higher likelihood. Normalise against the uncertainty-corridor radius, not a fixed distance.

Evidence signal	Why it moves the score
Time consistency	Vessel track intersects the plausible origin time window (from drift backtracking) → higher likelihood. A vessel only present hours after the estimated origin time is weak evidence.
Heading compatibility	Vessel heading around the origin time is consistent with a plausible spill/drift direction → higher likelihood.
Speed consistency	Speed profile plausible for normal transit near the source (very high speed sometimes correlates with evasive movement, but this is weak/soft evidence, not proof).
Vessel type	Tankers and vessels known to carry bulk oil/fuel get contextual weighting up; this must stay a soft multiplier, not a hard filter — non-tankers can still spill bunker fuel.
Track continuity	A continuous AIS track through the region is stronger evidence than a track with large gaps (gaps can also indicate AIS spoofing/switch-off, which is itself a soft red flag).
Environmental consistency	Candidate location falls inside the drift-backtracked uncertainty corridor → higher likelihood; outside it → sharply reduced.

3. Attribution Model — Options Compared

Approach	Explainability	Build difficulty	What it does	Verdict
Weighted scoring	High	Low (2-3/10)	Simple linear combination of normalised evidence features	Recommended baseline
Bayesian / likelihood model	Medium-High	Medium (5-6/10)	Treats each evidence type as updating a posterior probability per vessel	Good stretch goal if baseline finishes early
Bayesian network / PGM	Medium	High (8/10)	Explicit dependency graph between evidence variables	Not realistic for this timeline
Temporal graph / GNN	Low (for judges)	Very high (9-10/10)	Learns attribution from trajectory graph structure	Needs training data we don't have; skip

Recommendation: weighted scoring, with a Bayesian refinement as an optional layer on top if time remains. The reasoning: a judge evaluating a hackathon prototype will ask why a specific vessel was flagged, and a weighted sum with visible per-feature contributions answers that in one sentence per feature. A Bayesian network is scientifically nicer but needs conditional-probability estimates we have no data to calibrate honestly in one day — using made-up priors would be worse for credibility than a transparent weighted model. Graph/GNN approaches need training data we don't have and can't be defended live.

4. Recommended Scoring Design

Each evidence signal from Section 2 is normalised to a 0-1 sub-score, then combined:

$$\text{score}(\text{vessel}) = w_d \cdot f_{\text{distance}} + w_t \cdot f_{\text{time}} + w_h \cdot f_{\text{heading}} + w_s \cdot f_{\text{speed}} + w_{\text{type}} \cdot f_{\text{type}} + w_c \cdot f_{\text{continuity}} + w_{\text{env}} \cdot f_{\text{env_consistency}}$$

Suggested starting weights (tune after seeing real candidate data from Jitaan):

Feature	Starting weight	Rationale
Environmental consistency (inside corridor)	0.25	Hard gate-like signal — outside the corridor should suppress the whole score
Distance to inferred origin	0.20	Core spatial evidence
Time consistency	0.20	Core temporal evidence
Track continuity	0.15	Penalises suspicious AIS gaps
Heading compatibility	0.10	Supporting motion evidence
Speed consistency	0.05	Weak/soft evidence only
Vessel type	0.05	Contextual nudge, never a hard filter

Output per vessel: a 0-100 confidence score, plus a short generated reason string built from whichever 1-2 features contributed most (e.g. *"Passed within 8 km of the backtracked source region 3 hours before detection"*) — this reason string is exactly what Rudra's dashboard needs to display next to each ranked vessel.

Guardrail for the pitch: always present this as a *probable-attribution* score, never as proof. Both Martin's and Rudra's research independently flag this same caution — SAR cannot confirm the substance is oil chemically, and AIS proximity does not establish legal responsibility. The output should read as investigative triage, not a verdict.

5. Novelty — What We Can Actually Claim

Existing operational systems already combine satellite detection with AIS-based vessel analysis, so "we combine satellites and AIS" is not a defensible novelty claim on its own.

Existing system/work	What it does	Gap we can target
CleanSeaNet (EMSA)	European operational service; SAR/optical detection plus AIS, weather and oceanographic data to flag possible source vessels/installations.	Not open, not India-focused, pipeline not public
Cerulean (SkyTruth + Global Fishing Watch)	Scans Sentinel-1 with a ResNet-34 U-Net, then scores candidate vessels using slick-path alignment, proximity and timing against AIS history.	Global tool, not tuned to Indian waters, not an open student dashboard
Multi-source ship-tracing research (2024 study)	Academic method combining satellite, AIS and marine-environment data to trace a spill back to a probable source vessel.	Research paper, not a deployable dashboard

Our defensible novelty claim:

A low-cost, India-focused, explainable dashboard that fuses Sentinel-1 oil-spill segmentation, AIS vessel tracks, and Indian Ocean environmental data (Copernicus Marine / INCOIS) into a transparent, per-feature-scored vessel ranking with visual drift backtracking — built and demoed end to end within a single hackathon cycle.

The defensible part is **explainability** (a visible per-feature score breakdown, not a black-box output) and **Indian-waters framing** (INCOIS/Arabian Sea/Bay of Bengal context), not the base idea of satellite+AIS fusion, which is already proven elsewhere. This framing survives a judge who has heard of CleanSeaNet or Cerulean, because we're not claiming to have invented the concept — we're claiming a specific, useful gap: an open, explainable, India-tuned version of it.

6. System Interfaces (What I Need From Each Module)

My module is purely an aggregator — it needs clean, minimal inputs from the other two modules:

Handoff	Fields needed
From Martin (satellite)	spill_id, polygon, centroid [lat,lon], area_km2, detection_timestamp, confidence
From Jitaan (AIS/backtracking)	per-vessel: mmsi, name, type, position, track[], distance_km, time_difference_hours, heading, speed; plus source_region {lat, lon, radius_km, backtrack_hours}
Output to Rudra (dashboard)	ranked vessel list: mmsi, name, type, confidence (0-100), reason string, per-feature sub-scores (for an optional "why this vessel" popup)

One-line architecture for the pitch deck:

Sentinel-1 SAR → U-Net spill mask → geometry → wind/current backtracking → origin corridor → AIS candidate shortlist → **weighted evidence scoring** → ranked, explainable vessel attribution → dashboard

This matches the end-to-end architecture already laid out in Martin's report (Section 19) and Person 4's suggested JSON contract — my scoring module is the layer that turns Jitaan's raw candidate list into the vessels[] array with confidence and reason fields that Rudra's dashboard already expects.

7. Relevant Prior Work (Attribution-Specific)

Source	Where	What it does	Why it matters to us
CleanSeaNet operational service	EMSA	Combines SAR/optical detection with AIS, weather and oceanographic data to flag possible source vessels	Confirms the fused-evidence approach is operationally trusted, not just academic
Automatic SAR-Based Oil Spill Detection and Its Application to CleanSeaNet	ScienceDirect	Uses SAR plus meteorological, oceanographic and AIS data to estimate whether a dark feature is oil and help identify its source	Strong precedent for combining exactly our evidence types
Cerulean methodology	SkyTruth / Global Fishing Watch	Sentinel-1 + U-Net detection; scores candidate vessels using slick-path alignment, proximity and timing	Closest existing analogue to our weighted-scoring approach — useful to cite and differentiate against
A New Ship Tracing Technology from Oil Spills Based on Multi-Source Data (2024)	PubMed (via team Rudra research)	Uses satellite, AIS and marine-environment data to trace a spill to a probable source vessel	Closest academic precedent for our exact pipeline shape

8. Technical Risks

Risk	Severity	Mitigation
Weights are guessed, not learned	Medium	State clearly in the pitch that weights are hand-set based on domain reasoning, not trained — this is honest and still defensible for a prototype
Garbage-in from upstream modules	High	Agree on the interface contract (Section 6) with Martin and Jitaan early tonight so integration isn't rushed tomorrow night
Overclaiming certainty to judges	High	Every score is presented as "probable attribution," never "the vessel responsible" — consistent with both Martin's and Rudra's caution notes
No real candidate data to test against until Jitaan's module is ready	Medium	Build the scoring function against a small hand-written mock vessel list first, matching Rudra's JSON schema, so it can be integrated the moment real data is available

9. Final Decisions Summary

Question	Decision
Attribution approach	Weighted multi-evidence scoring (Bayesian refinement optional stretch)
Evidence signals used	Distance, time consistency, heading, speed, vessel type, track continuity, environmental consistency
Output format	0-100 confidence score + short reason string + per-feature sub-scores
Novelty claim	Explainable, India-focused, open evidence-fusion dashboard — not "first to combine satellite + AIS"
Closest existing systems	CleanSeaNet (EMSA), Cerulean (SkyTruth)
Main guardrail	Present as probabilistic triage, never as proof of responsibility

10. References and Verified Links

Sources drawn from the team's already-verified research (Martin's satellite report and Rudra's environmental/prior-art report) plus this module's own framing.

- [1] SIH26143 problem description — <https://sih26ps.vercel.app/>
- [2] EMSA — How does SAR detection work — <https://www.emsa.europa.eu/earth-observation-products/how-does-sar-detection-work.html>
- [3] EMSA CleanSeaNet overview — <https://www.emsa.europa.eu/csn-menu.html>
- [4] CleanSeaNet publication (EMSA) — <https://www.emsa.europa.eu/publications/item/4322-cleanseanet-european-satellite-based-oil-spill-monitoring-and-vessel-detection-service.html>
- [5] Automatic SAR-Based Oil Spill Detection and Its Application to CleanSeaNet — <https://www.sciencedirect.com/science/article/abs/pii/S0025326X13002695>
- [6] SkyTruth Cerulean — methods — <https://skytruth.org/cerulean/methods>
- [7] SkyTruth Cerulean — project overview — <https://skytruth.org/cerulean>
- [8] A New Ship Tracing Technology from Oil Spills Based on Multi-Source Data (2024, PubMed) — link on file in team's Rudra research report
- [9] GSCAT-UNET: Enhanced U-Net with Spatial-Channel Attention for Oil Spill Detection (2025) — <https://pubmed.ncbi.nlm.nih.gov/39862681/>
- [10] Martin's SIH26143 Satellite Oil-Spill Detection research report — Section 17-19 (backtracking, AIS correlation, end-to-end architecture) — team internal document
- [11] Rudra's SIH26143 Complete Research and Demo Plan — Parts C-E (prior art, existing systems, dashboard JSON contract) — team internal document